

LAB REPORT 01

INSTALLATION AND FIRST C++ PROGRAM

1. OBJECTIVE

- To download and install DEV-C++ or VS Code IDE.
- To open and explore the programming environment.
- To create a new C++ source file.
- To write, compile, and run a basic C++ Hello World program.
- To create a GitHub account and manage code repositories.

2. THEORETICAL BACKGROUND

Integrated Development Environment (IDE). An IDE like DEV-C++ or Visual Studio Code provides a comprehensive environment for writing, compiling, and debugging code. It integrates a code editor with a compiler (like MinGW) to translate C++ code into machine-executable programs.

Basic Program Structure. Every C++ program must have a `main()` function, which serves as the entry point for execution. The `#include` directive is used to include standard input/output stream libraries, enabling the program to print text to the console.

Version Control with GitHub. GitHub is a web-based platform that uses Git for version control. It allows developers to store their code in repositories, track changes, and share their work. Making a repository for lab tasks is a best practice for managing academic programming assignments.

3. SOFTWARE USED

- IDE: DEV-C++ IDE / VS Code
- Compiler: MinGW C++ Compiler
- Version Control: Git & GitHub

4. EXPERIMENTAL TASKS

Task 1: First C++ Program (Hello World)

Write, compile, and run a basic C++ program that prints 'Hello World!' to the console output window to verify that the IDE and compiler are configured correctly.

```
#include <iostream>
using namespace std;

int main() {
    // Print a simple message to the console
    cout << "Hello World!";
    return 0;
}
```

5. CONCLUSION

This introductory lab successfully familiarized the basic tools required for C++ programming. By installing the DEV-C++ IDE and writing the foundational 'Hello World!' program, the workflow of creating, compiling, and executing source code was established. Furthermore, exploring GitHub for version control introduced industry-standard practices for code management and sharing.